Type MS · Area of a triangle or quadrilateral from coordinates

Answer 1 of 17 · Question 6 part c · 4 marks

2020 JAN · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

Number		
6 (a)	$\frac{\text{Change in } y}{\text{Change in } x} = \frac{2-0}{2-3} = -2$	M1 A1
	$2 = 1 + c$	M1
	$c = 1$	A1 ft
	$x - 2y + 2 = 0$	A1
		(5)
(b)	$2y - 2 = 7y + 3$	M1
	$-5y = 5$	
	$y = -1$	A1
	When $y = -1$	
	$x = 2 \times -1 - 2 = -4$ So $C = (-4, -1)$	A1
	$\left(\frac{3-4}{2}, \frac{0-1}{2}\right) = \left(-\frac{1}{2}, -\frac{1}{2}\right)$	M1A1
		(5)
(c)	$AB = \sqrt{5}$	M1
	$BC = \sqrt{45}$	A1
	Area $= \frac{1}{2} \times \sqrt{5} \times \sqrt{45} \times \frac{1}{2}$	M1
	3.75	A1
		(4)
	Alternative c	**(14)**
	$\pm\frac{1}{2}\begin{vmatrix} 3 & 0 & 1 \\ 2 & 2 & 1 \\ -4 & -1 & 1 \end{vmatrix}$	M1
	$\pm\frac{1}{2}\left[3\begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} + 1\begin{vmatrix} 2 & 2 \\ -4 & -1 \end{vmatrix}\right]$	A1
	$\pm\frac{1}{2}[3(2+1) + (-2+8)] \times \frac{1}{2}$	M1
	3.75	A1

Part	Mark	Additional Guidance
(a)	M1	For an attempt to find the gradient using the given coordinates **and** a correct attempt to find the perpendicular gradient. Accept either $\frac{2-0}{2-3} = (-2)$ or $\frac{0-2}{3-2} = (-2) \Rightarrow m_p = -\frac{1}{'-2'}$
	A1	For $m = \frac{1}{2}$
	dM1	For a correct method to find the equation of a line $y - 2 = '\frac{1}{2}'(x-2)$ or $y - 0 = '\frac{1}{2}'(x-3)$ **The gradient must come from a correct attempt to find the gradient and the gradient of the perpendicular** If $y = mx + c$ is used, then they must use the correct values of x and y and a value for c must be reached before this mark is awarded.
	A1	For the correct equation in any form $y - 2 = \frac{1}{2}(x-2)$ or $y - 0 = \frac{1}{2}(x-3)$ or $y = \frac{1}{2}x + 1$ oe

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 1 of 17 · Question 6 part c · 4 marks

2020 JAN · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

	A1	For the correct equation in the required form $x-2y+2=0$ **oe** arranged in any order but all one side (e.g. accept even $\frac{x}{2}-y+1=0$)	
(b)	M1	Sets $L_1=L_2$ **and** attempts to solve for y or x	
		$2y-2=7y+3$ $-5y=5\Rightarrow y=...$	$\frac{x+2}{2}=\frac{x-3}{7}\Rightarrow 7x+14=2x-6$ $5x=-20\Rightarrow x=...$
	A1	$y=-1$	$x=-4$
	A1	$x=-4$	$y=-1$
	M1	For any **correct** method to find the coords of M using **their** values for C of x and y and the given coordinates of A (3, 0) $\left(\frac{3+['-4']}{2},\frac{0+['-1']}{2}\right)$ **This is a B mark in Epen**	
	A1	$\left(-\frac{1}{2},-\frac{1}{2}\right)$ **This is a B mark in Epen**	
(c)	M1	For attempting to find the length AB **and** BC $AB=\sqrt{(3-2)^2+(0-2)^2}$ and $BC=\sqrt{(2--4)^2+(2--1)^2}$ **This is a B mark in Epen**	
	A1	For both $AB=\sqrt{5}$ and $BC=\sqrt{45}$ **This is a B mark in Epen**	
	M1	$\frac{1}{2}('\sqrt{5}'\times'\sqrt{45}')\times\frac{1}{2}$	For using a correct method to find the area of the triangle using correct lengths. i.e. they must be using BC and AB
	A1	For $A=3.75$	
	ALT – using determinants		
	M1	For using a correct method with their coordinates for C in any order (it is a triangle), but they must start and finish with the same coordinates $A=\frac{1}{2}\begin{pmatrix}3 & 2 & '-\frac{1}{2}' & 3\\ 0 & 2 & '-\frac{1}{2}' & 0\end{pmatrix}$ **This is a B mark in Epen**	
	A1	For using the correct coordinates $A=\frac{1}{2}\begin{pmatrix}3 & 2 & -\frac{1}{2} & 3\\ 0 & 2 & -\frac{1}{2} & 0\end{pmatrix}$ **This is a B mark in Epen**	
	M1	For a correct evaluation using their coordinates $A=\frac{1}{2}\left(\left[3\times2+2\times'-\frac{1}{2}'+'-\frac{1}{2}'\times0\right]-\left[2\times0+'-\frac{1}{2}'\times2+3\times'-\frac{1}{2}'\right]\right)=...$	

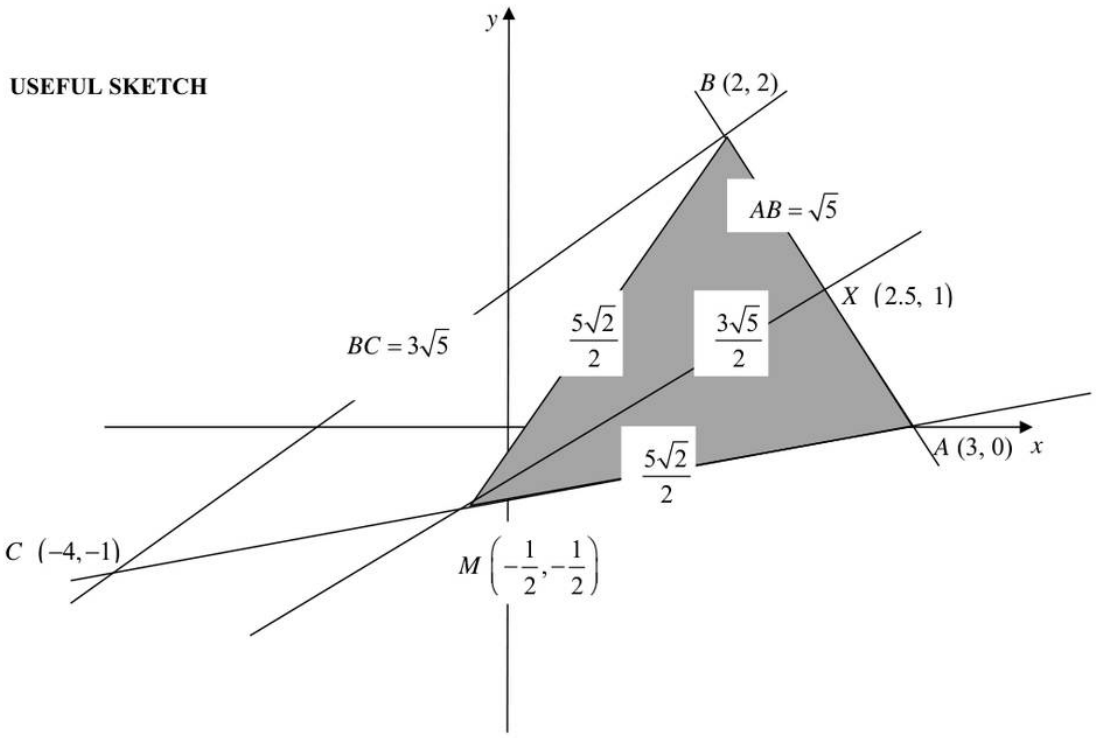
Type MS · Area of a triangle or quadrilateral from coordinates

Answer 1 of 17 · Question 6 part c · 4 marks

2020 JAN · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

	A1	For $A = 3.75$
	ALT	
	M1	For finding the length $AB = \sqrt{(3-2)^2 + (0-2)^2}$ **and** $MX = \frac{1}{2}\sqrt{3^2 + \left(\frac{3}{2}\right)^2}$ (Let X be midpoint of AB so MX is height of triangle ABM)
	A1	$AB = \sqrt{5}$ $\quad MX = \frac{3\sqrt{5}}{2}$
	M1	Area of $\Delta ABM = \frac{1}{2} \times AB \times MX = \frac{1}{2} \times '\sqrt{5}' \times '\frac{3\sqrt{5}}{2}' = \left(\frac{15}{4}\right)$
	A1	Area of $\Delta ABM = \frac{15}{4} = 3.75$
If they use trigonometry, please send to review		

USEFUL SKETCH



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 1 of 17 · Question 6 part c · 4 marks

2020 JAN · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 2 of 17 · Question 11 part e · 4 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 4 · find part e in this mark scheme extract

Number		
11 (a)	$\frac{y-6}{x-5}=\frac{6-3}{5--1}\Rightarrow 2y-12=x-5$	M1A1 [2]
(b)	$\left(\frac{2\times5+1\times-1}{2+1},\frac{2\times6+1\times3}{3}\right)\Rightarrow(3,\ 5)$*	M1A1 [2]
(a)		
M1	A fully correct method for finding the equation of a straight line e.g. $\frac{y-y_1}{x-x_1}=\frac{y_2-y_1}{x_2-x_1}$	
A1	$2y-12=x-5$ oe	
(b)		
M1	Use of $\left(\frac{qx_1+px_2}{p+q},\frac{qy_1+py_2}{p+q}\right)$ or $\begin{pmatrix}-1\\3\end{pmatrix}+\frac{2}{3}\times\begin{pmatrix}6\\3\end{pmatrix}$	
A1	Obtains the **given** coordinates	

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 2 of 17 · Question 11 part e · 4 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 4 · find part e in this mark scheme extract

(c)	Gradient of perpendicular to AB = -2	B1
	$\frac{5-n}{3-m}=-2\Rightarrow(11=2m+n)$	M1
	Radius = 5, so the length of AC = 10	
	$100=(3-n)^2+(-1-m)^2\Rightarrow(90=n^2-6n+m^2+2m)$	M1
	Solves simultaneous equations	
	$90=121-44m+4m^2-6(11-2m)+m^2+2m$	depM1
	$35=5m^2-30m\Rightarrow m^2-6m-7=0$	depM1
	$(m-7)(m+1)=0\Rightarrow m=7,\ n=-3$	A1
		[6]
ALT (c) using vectors		
(c)	$\overrightarrow{AB}=\begin{pmatrix}5\\6\end{pmatrix}-\begin{pmatrix}-1\\3\end{pmatrix}=\begin{pmatrix}6\\3\end{pmatrix}\Rightarrow$ Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$	B1
	$\lvert\overrightarrow{AB}\rvert=\sqrt{45}=3\sqrt{5}\Rightarrow\lvert AP\rvert=2\sqrt{5}\qquad\lvert\overrightarrow{AC}\rvert=10$	M1M1
	Using Pythagoras $\lvert\overrightarrow{PC}\rvert=\sqrt{100-20}=\sqrt{80}=4\sqrt{5}$	depM1
	$\overrightarrow{PC}=\frac{4\sqrt{5}}{3\sqrt{5}}\times\begin{pmatrix}3\\-6\end{pmatrix}=\begin{pmatrix}4\\-8\end{pmatrix}$	depM1
	$\Rightarrow$ Coordinates of C are $(3+(4),\ 5+(-8))\Rightarrow(7,-3)$	A1
		[6]

(c)

B1 Gradient of perpendicular to AB = -2 (Can be implied by M1)

M1 Obtains an equation using the gradient of the perpendicular and the points (m, n) and P (Condone if given in terms of x and y)

M1 Obtains a second equation using AC = 10 and the points A and C

depM1 Solve simultaneously to obtain a 3 term quadratic

depM1 Solve their 3 term quadratic

A1 $m = 7,\ n = -3$

ALT:

B1 Perpendicular vector to $\overrightarrow{AB}=\begin{pmatrix}3\\-6\end{pmatrix}$

M1 Finds $\lvert\overrightarrow{AB}\rvert$ or $\lvert\overrightarrow{AP}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\lambda\begin{pmatrix}3\\-6\end{pmatrix}$

M1 States $\lvert\overrightarrow{AC}\rvert=10$ or $\lambda=\frac{4}{3}$

depM1 Use of Pythagoras to find $\lvert\overrightarrow{PC}\rvert$ or $\overrightarrow{PC}=\begin{pmatrix}3\\5\end{pmatrix}+\frac{4}{3}\times\begin{pmatrix}3\\-6\end{pmatrix}$

depM1 Finds $\overrightarrow{PC}$

A1 $m = 7,\ n = -3$

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 2 of 17 · Question 11 part e · 4 marks

2019 JUN · Paper 01 (regional R) · extract 3 of 4 · find part e in this mark scheme extract

(d)	$\frac{y-'-3'}{x-'7'}=\frac{1}{2}\Rightarrow\left\{y=\frac{x-13}{2}\right\}$, $\frac{y-3}{x--1}=-2\Rightarrow\{y=-2x+1\}$ Solving simultaneous equations by any method $\frac{x-13}{2}=-2x+1\Rightarrow p=3$ and $q=-5$	M1 M1A1 [3]
ALT (d) using vectors		
(d)	$\overrightarrow{AD}=\overrightarrow{PC}=\begin{pmatrix}4\\-8\end{pmatrix}$ Coordinates of point D $\Rightarrow(-1+(4),\ 3+(-8))=(3,-5)$	M1 M1A1 [3]
(e)	Length of AB $\sqrt{(6-3)^2+(5--1)^2}=3\sqrt{5}$ Length of CD $\sqrt{(-3--5)^2+(7-3)^2}=\sqrt{20}=2\sqrt{5}$ Area of trapezium $=\frac{1}{2}(3\sqrt{5}+2\sqrt{5})\times 4\sqrt{5}=50$ (units^2)	M1 A1 M1A1 [4]
ALT (e) using vectors		
	Area $=\frac{1}{2}\begin{vmatrix}-1 & 5 & 7 & 3 & -1\\ 3 & 6 & -3 & -5 & 3\end{vmatrix}$ $=\frac{1}{2}[(-6-15-35+9)-(15+42-9+5)]=50$	 M1A1 M1A1 [4]
		Total 17 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 2 of 17 · Question 11 part e · 4 marks

2019 JUN · Paper 01 (regional R) · extract 4 of 4 · find part e in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 3 of 17 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 1 of 4 · find part c in this mark scheme extract

5(a)	Gradient $PR = \frac{6}{-6} = -1$, Gradient $QS = \frac{8}{8} = 1$	M1A1
	Product $= -1 \Rightarrow$ perpendicular	A1 (3)
(b)	(i) $PR = \sqrt{6^2+6^2} = 6\sqrt{2}\ \left(=\sqrt{72}\right)$	M1
	(ii) $QS = \sqrt{8^2+8^2} = 8\sqrt{2}\ \left(=\sqrt{128}\right)$	A1 (2)
(c)	Area $= \frac{1}{2}\text{"}6\sqrt{2}\text{"}\times\text{"}8\sqrt{2}\text{"} = 48\ \left(\text{units}^2\right)$	M1A1 (2)
		[7]

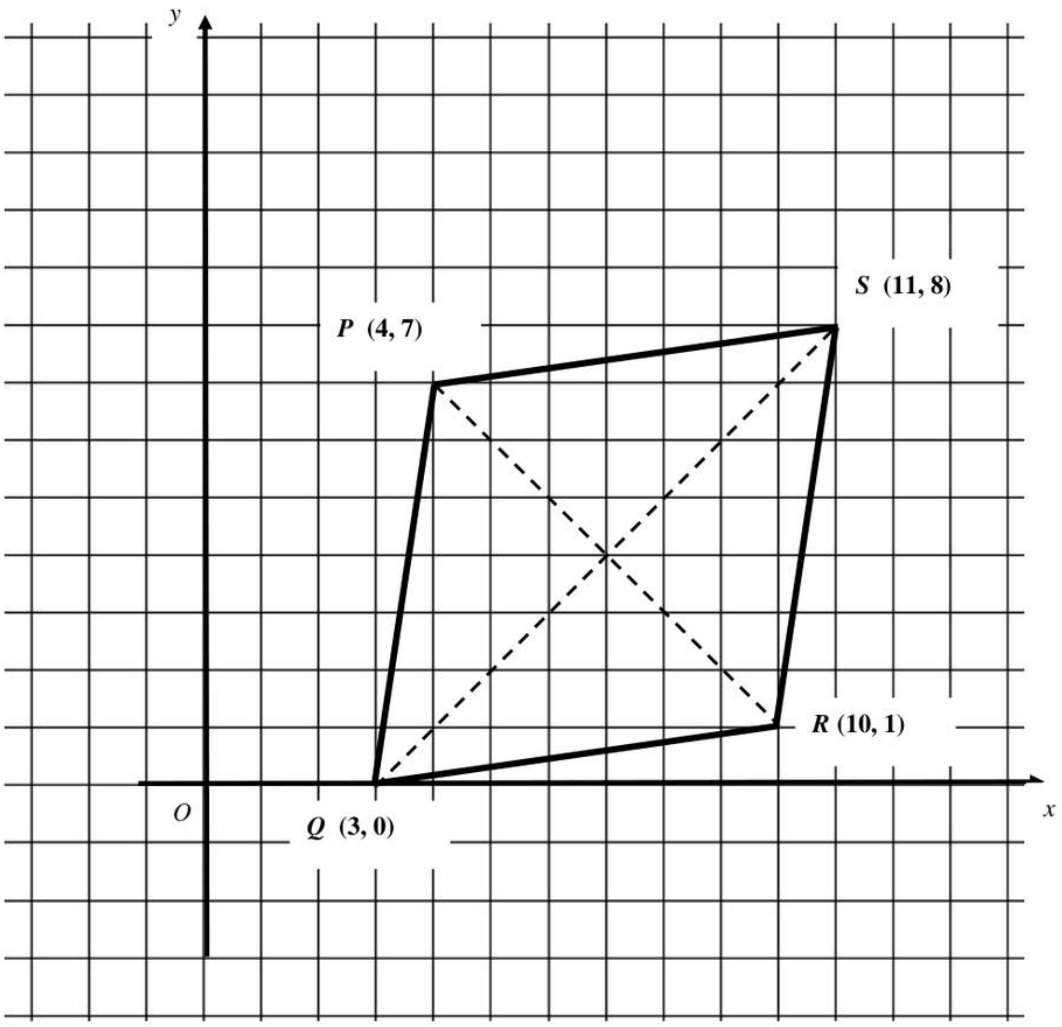
Part	Mark	Notes
(a)	M1	Finds the gradient of PR and QS using a correct method. This may be on a diagram. Gradient $PR = \frac{7-1}{4-10} = \frac{6}{-6} = -1$, Gradient $QS = \frac{8-0}{11-3} = \frac{8}{8} = 1$
	A1	Both gradients correct Gradient $PR = -1$, Gradient $QS = 1$
	A1	Finds the product of the two gradients with a statements that as the product = –1 then the lines are perpendicular.
(b)	M1	For **either** $PR = \sqrt{6^2+6^2} = \sqrt{72}$ or $6\sqrt{2}$ **OR** $QS = \sqrt{8^2+8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
	A1	For **both** $PR = \sqrt{6^2+6^2} = \sqrt{72}$ or $6\sqrt{2}$ **AND** $QS = \sqrt{8^2+8^2} = \sqrt{218}$ or $8\sqrt{2}$ correct
(c)	M1	$PQRS$ is a kite so $= \frac{1}{2}\text{"}6\sqrt{2}\text{"}\times\text{"}8\sqrt{2}\text{"} = \ldots$
	A1	Area = $48\ \left(\text{units}^2\right)$
	ALT Uses determinants	
	M1	$\text{Area} = \frac{1}{2}\begin{pmatrix} 4 & 3 & 10 & 11 & 4 \\ 7 & 0 & 1 & 8 & 7 \end{pmatrix}$ $= \frac{1}{2}\left(\left[4\times0+3\times1+10\times8+11\times7\right]-\left[3\times7+10\times0+11\times1+4\times8\right]\right)$ $=\ldots$ Allow one slip in a product.
	A1	Area = $48\ \left(\text{units}^2\right)$

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 3 of 17 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 2 of 4 · find part c in this mark scheme extract

USEFUL SKETCH



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 3 of 17 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 3 of 4 · find part c in this mark scheme extract

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 3 of 17 · Question 5 part c · 2 marks

2020 NOV · Paper 01 · extract 4 of 4 · find part c in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 4 of 17 · Question 10 part d · 3 marks

2022 JAN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

10 (a)	For the correct value of $a = 10$	B1 [1]
(b)	Gradient of line L_2 $\quad m = -\frac{1}{2}$ $y - '10' = '-\frac{1}{2}'(x-2)$ $y - 10 = -\frac{1}{2}(x-2)$ oe $x + 2y - 22 = 0$ *	B1 M1 A1 A1 cso [4]
(c)	Coordinates of point A are $(-3,\ 0)$ Coordinates of point B are $(22,\ 0)$ Length of AC $(5\sqrt{2})^2 = (m - '-3')^2 + n^2 \quad [\Rightarrow 50 = m^2 + 6m + 9 + n^2]$ Gradient of BC $\frac{1}{4} = \frac{n}{m-'22'} \Rightarrow n = \frac{m-'22'}{4} \Rightarrow n^2 = \frac{m^2-44m+484}{16}$ $50 = m^2 + 6m + 9 + \frac{m^2 - 44m + 484}{16}$ [or $50 = (22+4n)^2 + 6(22+4n) + 9 + n^2$] $17m^2 + 52m - 172 = 0$ OR $17n^2 + 200n + 575 = 0$ e.g. $m = \frac{-52 \pm \sqrt{52^2 - 4\times17\times-172}}{2\times17} \Rightarrow m = 2,\ (-5.058...)$ $m = 2$ and $n = -5$	B1 B1 M1 M1 ddM1 A1 M1 A1A1 [9]
(d)	Area$_{APB} = \frac{1}{2}('10') \times ('22'-'3') = (125)$ Area$_{ABC} = \frac{1}{2}('5') \times ('22'-'-3') = \left(\frac{125}{2}\right)$ Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2	M1 A1 A1 [3]
	ALT	
	Area$_{ACBP} = \frac{1}{2}('25') \times ('15') = ...$ Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2	[M1A1 A1]
	ALT	

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 4 of 17 · Question 10 part d · 3 marks

2022 JAN · Paper 01 · extract 2 of 4 · find part d in this mark scheme extract

	$A=\frac{1}{2}\begin{bmatrix}'-3' & 2 & '22' & '2' & '-3'\\ 0 & '10' & 0 & '-5' & '0'\end{bmatrix}$	[M1
	$A=\frac{1}{2}\left(\left[(-3)\times10+2\times0+22\times(-5)+2\times0\right]-\left[2\times0+22\times10+2\times0+(-3)\times-5\right]\right)=...$	A1
	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2	A1]
		Total 17 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 4 of 17 · Question 10 part d · 3 marks

2022 JAN · Paper 01 · extract 3 of 4 · find part d in this mark scheme extract

		Any other values of m and n must be rejected.
(d)	**M1**	For either area of triangle APB or ABC $\text{Area}_{APB} = \frac{1}{2}('10')\times('22'--'3') = (125)$ or $\text{Area}_{ABC} = \frac{1}{2}('5')\times('22'-'-3') = \left(\frac{125}{2}\right)$
	A1	Either area of triangle APB or ABC correct
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT		
	M1A1	$\text{Area}_{ACBP} = \frac{1}{2}('25')\times('15') = \ldots$
	A1	Area of quadrilateral $ACBP = \frac{375}{2}$ (units)^2
ALT – determinant method		
	M1	For using the 'determinant method' e.g. $A = \frac{1}{2}\begin{bmatrix} '-3' & 2 & '22' & '2' & '-3' \\ 0 & '10' & 0 & '-5' & '0' \end{bmatrix}$
	A1	For a correct evaluation of their determinants using **their** values e.g. $A = \frac{1}{2}\left(\left[(-3)\times10+2\times0+22\times(-5)+2\times0\right]-\left[2\times0+22\times10+2\times0+(-3)\times-5\right]\right) = \ldots$
	A1	Area of quadrilateral $APBC = \frac{375}{2}$ (units)^2

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 4 of 17 · Question 10 part d · 3 marks

2022 JAN · Paper 01 · extract 4 of 4 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 5 of 17 · Question 6 part d · 3 marks

2020 NOV · Paper 02 (regional R) · extract 1 of 6 · find part d in this mark scheme extract

6 (a)	$x^2-5x+4=(x-4)(x-1)$	M1
	$A(1,0)$ $B(4,0)$	A1 A1 (3)
	Notes	
(a)		
M1	For solving the quadratic	
A1	For $A(1,0)$	
A1	For $B(4,0)$	
(b)	$\frac{dy}{dx}=2x-5$	M1
	When $x=1$ $\frac{dy}{dx}=-3$	A1
	Tangents meet on the axis of symmetry of curve C so $x=\frac{1+4}{2}=\frac{5}{2}$	M1 A1
	When $x=\frac{5}{2}$ $y=-3\left(\frac{5}{2}-1\right)=-\frac{9}{2}$	M1
	$\left(\frac{5}{2},-\frac{9}{2}\right)$	A1 (6)
	Notes	
(b)		
M1	For $\frac{dy}{dx}=2x-5$ (Allow if seen in part(c))	
A1	$\frac{dy}{dx}=-3$ when $x=1$	
M1	For use of $\frac{x_1+x_2}{2}$ or $\frac{-b}{2a}$ or $\frac{dy}{dx}=0$	
A1	$x=\frac{5}{2}$	
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m(x-x_1)$ oe	
A1	For $\left(\frac{5}{2},-\frac{9}{2}\right)$	

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 5 of 17 · Question 6 part d · 3 marks

2020 NOV · Paper 02 (regional R) · extract 2 of 6 · find part d in this mark scheme extract

	Alternative Method $\frac{dy}{dx} = 2x - 5$ When $x = 1$ $\frac{dy}{dx} = -3$ So $y = -3x + 3$ When $x = 4$ $\frac{dy}{dx} = 3$ So $y = 3x - 12$ Meet when $-3x + 3 = 3x - 12$ $\left(\frac{5}{2}, -\frac{9}{2}\right)$	M1 A1 M1 A1 M1 A1 (6)
	Notes	
	Alternative	
M1	For an attempt to find the equation of the tangent of C at A	
A1	For $y = -3x + 3$	
M1	For an attempt to find the equation of the tangent of C at B	
A1	For $y = 3x - 12$	
M1	For equating the two equations	
A1	For $\left(\frac{5}{2}, -\frac{9}{2}\right)$	

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 5 of 17 · Question 6 part d · 3 marks

2020 NOV · Paper 02 (regional R) · extract 3 of 6 · find part d in this mark scheme extract

(c)	Gradient of normal at (0, 1) = $\frac{1}{3}$	M1
	When $x=\frac{5}{2}$ $y=\frac{1}{3}\left(\frac{5}{2}-1\right)=\frac{1}{2}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)
	Alternative Method $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$	M1
	Meet when $\frac{1}{3}x-\frac{1}{3}=-\frac{1}{3}x+\frac{4}{3}$	M1
	$\left(\frac{5}{2},\frac{1}{2}\right)$	A1 (3)

	Notes
(c)	
M1	For Gradient of normal at (0, 1) = $\frac{1}{3}$ ft the gradient found in part (b)
M1	For substitution of $x=\frac{5}{2}$ into $y-y_1=m\left(x-x_1\right)$ oe
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$
	Alternative
M1	For $y=\frac{1}{3}x-\frac{1}{3}$ and $y=-\frac{1}{3}x+\frac{4}{3}$
M1	For equating the two equations
A1	For $\left(\frac{5}{2},\frac{1}{2}\right)$

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 5 of 17 · Question 6 part d · 3 marks

2020 NOV · Paper 02 (regional R) · extract 4 of 6 · find part d in this mark scheme extract

(d)	Area $= \frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 1 Area $= \frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}=\frac{15}{2}$	M1 M1 A1 (3)
	Alternative Method 2 $AN=\sqrt{1.5^2+0.5^2}=\frac{\sqrt{10}}{2}\quad AT=\sqrt{1.5^2+4.5^2}=\frac{3\sqrt{10}}{2}$	M1
	Area $= 2\times\frac{1}{2}\times\frac{\sqrt{10}}{2}\times\frac{3\sqrt{10}}{2}=\frac{30}{4}=\frac{15}{2}$	M1 A1 (3)
	Alternative Method 3 Area $= \frac{1}{2}\begin{vmatrix}1 & \frac{5}{2} & 4 & \frac{5}{2} & 1\\ 0 & -\frac{9}{2} & 0 & \frac{1}{2} & 0\end{vmatrix}\Rightarrow\frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)=\frac{15}{2}$	M1 M1 A1 (3)
		[15]

	Notes
(d)	
M1	For $\frac{1}{2}\times AB\times NT$
M1	For $\frac{1}{2}(4-1)\times\left(\frac{1}{2}+\frac{9}{2}\right)$
A1	For $\frac{15}{2}$
	Alternative 1
M1	For area of triangle ANB + area of triangle ATB
M1	For $\frac{1}{2}\times 3\times\frac{1}{2}+\frac{1}{2}\times 3\times\frac{9}{2}$
A1	$\frac{15}{2}$
	Alternative 2
M1	For finding AN and AT
M1	For $2\times\frac{1}{2}\times AN\times AT$
A1	$\frac{15}{2}$

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 5 of 17 · Question 6 part d · 3 marks

2020 NOV · Paper 02 (regional R) · extract 5 of 6 · find part d in this mark scheme extract

	Alternative 3
M1	Use of area $= \frac{1}{2}\begin{vmatrix} 1 & a & 4 & c & 1 \\ 0 & b & 0 & d & 0 \end{vmatrix}$ oe where (a, b) and (c, d) are the coordinates of T and N
M1	$\frac{1}{2}\left(-\frac{9}{2}+2+18-\frac{1}{2}\right)$
A1	$\frac{15}{2}$

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 5 of 17 · Question 6 part d · 3 marks

2020 NOV · Paper 02 (regional R) · extract 6 of 6 · find part d in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 6 of 17 · Question 6 part c · 2 marks

2022 JAN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

Number		
6 (a)	$\frac{dy}{dx} = \frac{2}{\sqrt{x}}$	M1
	When $x = 9$ $\frac{dy}{dx} = \frac{2}{3}$	A1
	$y - 12 = \frac{2}{3}(x - 9)$	M1
	When $y = 0$ $-12 = \frac{2}{3}(x - 9)$	M1
	$x = -9$ So $(-9, 0)$	A1 (5)
(b)	Gradient of Normal: $-\frac{3}{2}$	M1
	$y - 12 = -\frac{3}{2}(x - 9)$	M1
	When $y = 0$ $-12 = -\frac{3}{2}(x - 9)$	M1
	$x = 17$ So $(17, 0)$	A1 (4)
(c)	$\frac{1}{2} \times 12 \times 26 = 156$	M1 A1 (2)
		Total 11 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 6 of 17 · Question 6 part c · 2 marks

2022 JAN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For an attempt to differentiate y wrt x. Accept $\frac{dy}{dx} = 2x^{-\frac{1}{2}}$ Allow as a minimum $\frac{dy}{dx} = kx^{-\frac{1}{2}}$ where k is a constant $\neq 0$
	A1	For $\frac{dy}{dx} = \frac{2}{3}$
	M1	For a complete method to find the equation of a straight line. If they use $y = mx + c$ then they must reach a value for c for this mark. Accept the equation of the line with their value of $\frac{dy}{dx}$ using the given coordinates.
	M1	For substitution of $y = 0$ to find a value for x
	A1	For $(-9, 0)$
(b)	M1	For gradient of Normal: $-\frac{3}{2}$ Which is the negative reciprocal of their gradient obtained in (a)
	M1	For a complete method to find the equation of a straight line with a gradient of $-\frac{3}{2}$ (ft their gradient of normal) If they use $y = mx + c$ then they must reach a value for c for this mark.
	M1	For substitution of $y = 0$ to find a value for x
	A1	For $(17, 0)$
(c)	M1	For any correct method for finding the area of a triangle. e.g., $A = \frac{1}{2} \times 12 \times ('17' - '-9') = 156$ OR $A = \frac{1}{2}\begin{bmatrix} 9 & -9 & 17 & 9 \\ 12 & 0 & 0 & 12 \end{bmatrix} = \frac{1}{2}[(0+0+204)-(0+0-108)] = 156$
	A1	For 156

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 6 of 17 · Question 6 part c · 2 marks

2022 JAN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 7 of 17 · Question 5 part d · 4 marks

2023 JAN · Paper 02 (regional R) · extract 1 of 3 · find part d in this mark scheme extract

5(a)	$-\frac{1}{12}=\frac{10-0}{p-123}$ $\left[\text{or } -\frac{1}{12}=\frac{0-10}{123-p}\right]$ $\Rightarrow p-123=-12\times10\Rightarrow p=3^*$	M1 A1 cso [2]
(b)	$y-10=-\frac{1}{12}(x-3)\Rightarrow 12y+x-123=0$	M1A1 [2]
(c)	$m_k=12$ $y-10=12(x-3)\Rightarrow y=12x-26$	B1 M1A1 [3]
(d)	At C: $y=12x-26$ when $y=0$, $x=\frac{26}{12}$ oe $\text{Area}_{ABC}=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=\frac{3625}{6}$ ALT $\text{Area}=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\ 10 & 0 & 0 & 10\end{vmatrix}=\frac{3625}{6}$	M1A1 M1A1 [M1A1] [4]
		Total 11 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 7 of 17 · Question 5 part d · 4 marks

2023 JAN · Paper 02 (regional R) · extract 2 of 3 · find part d in this mark scheme extract

Question	Notes	Marks
5(a)	States a correct expression for the gradient in terms of p either $-\frac{1}{12}=\frac{10-0}{p-123}$ or $-\frac{1}{12}=\frac{0-10}{123-p}$ and attempts to solve their equation in p $p-123=-12\times10\Rightarrow p=...$	M1
	Finds the value of $p=3$* Must show an intermediate step e.g. $k=p\pm c$ or $c=\frac{p}{k}$ or $c=kp$	A1 cso [2]
(b)	Forms an equation using either the formula or $y=mx+c$ with the given values $y-10=-\frac{1}{12}(x-3)$ OR $y-0=-\frac{1}{12}(x-123)$	M1
	For the correct equation in the required form $12y+x-123=0$ Coefficients must be integers. Allow for $r=1, s=12, t=-123$	A1 [2]
(c)	For the gradient of the normal when $x=3$ is 12	B1
	Forms an equation of the normal using either the formula or $y=mx+c$ with their values of the gradient of the normal. $y-12=12(x-3)$	M1
	For the correct equation in the required form. $y=12x-26$	A1 [3]
(d)	Attempts to find the x coordinate at point C $y=12x-26$ when $y=0$, $x=...$ Their linear equation from (c) with $y=0$ and attempt to solve.	M1
	For the correct value of x $x=\frac{26}{12}$ oe	A1
	For attempting to find the area of triangle ABC Area $=\frac{1}{2}\times10\times\left(123-\frac{13}{6}\right)=...$ or Area $=\frac{1}{2}\begin{vmatrix}3 & 123 & 13/6 & 3\\10 & 0 & 0 & 10\end{vmatrix}$	M1
	For the correct area of the triangle $\frac{3625}{6}$ oe isw rounding	A1 [4]
		Total 11 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 7 of 17 · Question 5 part d · 4 marks

2023 JAN · Paper 02 (regional R) · extract 3 of 3 · find part d in this mark scheme extract

Question	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 8 of 17 · Question 9 part d · 2 marks

2024 JUN · Paper 01 (regional R) · find part d in this mark scheme extract

9(a)	$\frac{y-8}{8-3}=\frac{x-6}{6--4}\Rightarrow\frac{y-8}{5}=\frac{x-6}{10}\Rightarrow\left[y-8=\frac{1}{2}(x-6)\right]$	M1A1 [2]
(b)	$C=\left(\frac{4\times6+1\times-4}{4+1},\frac{4\times8+1\times3}{4+1}\right)=(4,7)$	B1,B1 [2]
(c)	$-2=\frac{q-7}{p-4}\Rightarrow q=15-2p$ $8\sqrt{5}=\sqrt{(4-p)^2+(7-q)^2}$ $\Rightarrow 320=(4-p)^2+(2p-8)^2\Rightarrow p^2-8p-48=0$ $\Rightarrow(p+4)(p-12)=0$ $\Rightarrow p=-4\quad q=15-2\times-4=23\Rightarrow D=(-4,23)$	M1 M1 M1 M1 A1A1 [6]
(d)	Area of triangle $ACD=\frac{1}{2}\times(23-3)\times(4--4)=80$ [square units] **ALT 1** $\frac{1}{2}\begin{bmatrix}-4 & 4 & -4 & -4\\ 3 & 7 & 23 & 3\end{bmatrix}=\frac{1}{2}\left[(-28+92-12)-(12-28-92)\right]=80$ **ALT 2** $A=\frac{1}{2}\times AC\times CD=\frac{1}{2}\times4\sqrt{5}\times8\sqrt{5}=80$	M1A1 [2] [M1A1] [M1A1]
		Total 12 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 9 of 17 · Question 2 part b · 5 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

Number		
2 (a)	$(-3,9)$	B1
	$(m_{PQ}=)\dfrac{18-3}{3--7}\left(=\dfrac{3}{2}\right)$ oe	M1
	$(m_l=)-\dfrac{1}{"\frac{18-3}{3--7}"}=\left(-\dfrac{1}{"\frac{3}{2}"}\right)\left(=-\dfrac{2}{3}\right)$ oe	M1
	$y-"9"="-\dfrac{2}{3}"(x+"3")$	M1
	$3y+2x-21=0$ oe eg $-6y-4x+42=0$	A1 [5]
(b)	$3\times0+2x-21=0\Rightarrow x=(10.5)$	M1
	$(QR^2=)(3-"-3")^2+(18-"9")^2\left(=117\right)$ or $(QR=)\sqrt{(3-"-3")^2+(18-"9")^2}\left(=3\sqrt{13}\right)$	M1
	$(RS^2=)("10.5"-"-3")^2+(0-"9")^2\left(=\dfrac{1054}{4}\right)$ or $(RS=)\sqrt{("10.5"-"-3")^2+(0-"9")^2}\left(=\dfrac{9\sqrt{13}}{2}\right)$	M1
	$\dfrac{1}{2}\times"3\sqrt{13}"\times"\dfrac{9\sqrt{13}}{2}"$	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1 [5]
ALT1	$3\times0+2x-21=0\Rightarrow x=(10.5)$	M1
	$\left(\dfrac{1}{2}\right)\begin{vmatrix}3 & "-3" & "10.5" & 3\\ 18 & "9" & 0 & 18\end{vmatrix}$ oe eg $\left(\dfrac{1}{2}\right)\begin{vmatrix}"10.5" & "-3" & 3 & "10.5"\\ 0 & "9" & 18 & 0\end{vmatrix}$ oe	M1 M1
	$\dfrac{1}{2}\left((3\times9+-3\times0+10.5\times18)-(3\times0+10.5\times9+-3\times8)\right)$ oe	ddM1
	$\dfrac{351}{4}$oe (87.75)	A1
	Finding the area of a rectangle and subtracting area of triangles	
ALT2a	$3\times0+2x-21=0\Rightarrow x=0$	M1
	$("10.5"-"-3")\times18=243$	M1
	$\dfrac{1}{2}\times(18-"9")\times(3-"-3")(=27)$	M1 M1
	$\dfrac{1}{2}\times("10.5"-3)\times(18(-0))\left(=\dfrac{135}{2}\right)$	
	$\dfrac{1}{2}\times("10.5"-"-3")\times("9"(-0))\left(=\dfrac{243}{4}\right)$	
	$243-"27"-"\dfrac{135}{2}"-"\dfrac{243}{4}"$	ddM1
	$\dfrac{351}{4}$ oe (87.75)	A1

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 9 of 17 · Question 2 part b · 5 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

ALT2b	Let M be the intersection of PQ with the x-axis	M1
	$3\times 0+2x-21=0\left(\Rightarrow x=10.5\right)$	M1
	$\left(y=\frac{3}{2}x+13.5\Rightarrow\ x=\right)-9$	M1
	$\left(\text{Area of triangle } SQM\ =\right)\frac{1}{2}\times\left("10.5"-"-9"\right)\times 18\left(=\frac{351}{2}\right)$	M1
	$\left(\text{Area of triangle } SRM\ =\right)\frac{1}{2}\times\left("10.5"-"-9"\right)\times 9\left(=\frac{351}{4}\right)$	ddM1
	$"\frac{351}{2}"-"\frac{351}{4}"$ $\frac{351}{4}$ oe (87.75)	A1
ALT3	Usually with a good diagram, the candidate can work out that triangles SRM and QRS are congruent (SAS) If spotting this, this is full marks or 0 marks (because without the correct coordinates, this is not so) $\frac{1}{2}\times 19.5\times 9=\frac{351}{4}$	M1M1 M1M1 ddM1A1
		Total 10 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 10 of 17 · Question 9 part c · 3 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

Number		
9 (a)	$a=-1,\ b=-2$	B1,B1 [2]
(b)	Gradient of $l_1=-2,\Rightarrow$ Gradient of $l_2=\frac{1}{2}$	B1,B1
	$180=(x+1)^2+(y-6)^2$	M1
	$\frac{1}{2}=\frac{y-6}{x+1}\Rightarrow x=2y-13$	M1
	Solves simultaneous equations; $180=([2y-13]+1)^2+(y-6)^2\Rightarrow 0=5y^2-60y$	M1M1
	or $180=(x+1)^2+\left(\frac{1}{2}x+\frac{13}{2}-6\right)^2\Rightarrow 0=x^2+2x-143=0$	
	$y=0,\ y=12\Rightarrow x=-13,\ x=11$ or $x=-13,\ x=11\Rightarrow y=0,\ y=12$ Coordinates are $(-13,\ 0)$ and $(11,\ 12)$	A1A1 [8]
(c)	Area of triangle PQR $PQ=\sqrt{(6+2)^2+(-1-3)^2}=4\sqrt{5}$	M1
	Area $=\frac{1}{2}\times 4\sqrt{5}\times 6\sqrt{5}=60$ (units)^2	M1A1 [3]
	ALT Area $=\frac{1}{2}\begin{pmatrix}-13 & -1 & 3 & -13\\ 0 & 6 & -2 & 0\end{pmatrix}=\frac{1}{2}(-78+2+0-0-18-26)=-60$	{M1}{M1}
	$\Rightarrow 60$ (units)^2	{A1} [3]
(d)	Coordinates of R required are $(-13,\ 0)$	
	$\angle RPQ=90^\circ$ so RQ is a diameter	
	$\left(\frac{-13+3}{2},\frac{0-2}{2}\right)\Rightarrow(-5,\ -1)$	M1A1 [2]
		Total 15 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 10 of 17 · Question 9 part c · 3 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

(b)	
B1	Gradient of $l_1 = -2$
B1	Gradient of $l_2 = \frac{1}{2}$
M1	Use of $PR = 6\sqrt{5}$ to obtain an equation
M1	Use of gradient of the perpendicular to obtain an equation
M1	Solves simultaneously
M1	Simplifies to $5y^2 - 60y = 0$ or $x^2 + 2x - 143 = 0$
A1	All 4 values identified 0, 12, –13, 11
A1	$(11,\ 12)$ and $(-13,\ 0)$ (must be paired correctly and if written as a coordinate then must be in the correct order)
(c)	
M1	$PQ = 4\sqrt{5}$
M1	Use of Area $= \frac{1}{2} \times PQ \times PR$
A1	60 (units)^2
ALT	
M1	Use of Area $= \frac{1}{2}\begin{pmatrix} -13 & -1 & 3 & -13 \\ 0 & 6 & -2 & 0 \end{pmatrix}$ ft R provided $e < 0$
M1	$\frac{1}{2}(-78 + 2 + 0 - 0 - 18 - 26)$ ft R provided $e < 0$
A1	60 (units)^2
(d)	
M1	$\left(\frac{-13+3}{2}, \frac{0-2}{2}\right)$ ft R provided $e < 0$
A1	$(-5,\ -1)$

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 10 of 17 · Question 9 part c · 3 marks

2020 JAN · Paper 01 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 11 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 5 · find part c in this mark scheme extract

11 (a)	$b = 33$	B1
	$\left(\left(\frac{3\times0+2\times10}{5}, \frac{3\times3+2\times33}{5}\right)\right) = (4, 15)$	B1 B1
	ALT	
	Identifies the correct x coordinate of $x = 4$ using any method.	B1
	$y = 3\times'4'+3 = ... \Rightarrow y = 15$	B1B1
	For the gradient of line k $\quad m = -\frac{1}{"3"}$	M1
	$y - "15" = -\frac{1}{"3"}(x - "4")$	dM1
	$3y + x - 49 = 0^*$ cso	A1* cso [6]
(b)	$A = (0, 3)$	B1
	$-\frac{1}{"3"} = \frac{q-"3"}{p(-0)} (\Rightarrow p = -"3"(q-"3"))$ **alt** $y = -\frac{1}{3}x + 3 \Rightarrow q = -\frac{p}{3} + 3$ **OR** $p^2 + (q-"3")^2 = 1440$	M1
	Equation in *q* $10q^2 - 60q + 90 = 1440 \Rightarrow q^2 - 6q - 135 = 0$ oe $10(q-3)^2 = 1440$ or $q^2 - 6q - 135 = 0$	dM1
	Equation in *p* $1440 = p^2 + \left(3 - 3 - \frac{p}{3}\right)^2 \Rightarrow 1440 = \frac{10p^2}{9}$ And attempting to solve by any valid method, reaching a value for p	
	$q - 3 = \pm\sqrt{144}$ or $(q-15)(q+9)$ or $\frac{--6 \pm \sqrt{(-6)^2 - 4.(1).(-135)}}{2\times1}$	ddM1
	$q = 15, p = -36$	A1 A1 [6]
	ALT	
	$A = (0, 3)$	[B1
	$a\sqrt{(-3)^2 + 1^2}$ $(=12\sqrt{10})$	M1
	$\Rightarrow a = (12)$	dM1
	$p = 0 - 3\times"12"$	dM1
	$q = 3 + 1\times"12"$	
	$p = -36$ and $q = 15$ or D is $(-36, 15)$	A1 A1]

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 11 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 5 · find part c in this mark scheme extract

(c)	**Finding the coordinates of points *E* and *F***	
	(Finding E $x = -36$ $3y + x - 49 = 0$)	M1
	$3y + ("-36") - 49 = 0 \left(y = \frac{85}{3} \right)$	
	(Finding F $x = -36$ $y - 3x = 3$)	M1
	$y - 3("-36") = 3 \ (y = -105)$	
	Find the area of triangle *ECF*	
	Method 1	
	$(CE =) \sqrt{("4" - "-36")^2 + \left("\frac{85}{3}" - "15" \right)^2} = \left(\frac{40\sqrt{10}}{3} \right)$	M1
	Or	
	$(CF =) \sqrt{("4" - "-36")^2 + ("15" - "-105")^2} = \left(40\sqrt{10} \right)$	
	$\left(\text{Area triangle } ECF = \right) \frac{1}{2} \times \frac{40\sqrt{10}}{3} \times 40\sqrt{10} = \frac{8000}{3}$ oe	M1A1 [5]
	Method 2	
	$\text{Area} = \frac{1}{2} \begin{bmatrix} "-36" & "4" & "-36" & "-36" \\ "\frac{85}{3}" & "15" & "-105" & "\frac{85}{3}" \end{bmatrix}$ oe	M1
	$\frac{1}{2} \left(\begin{bmatrix} "-36" \times "15" + "4" \times "-105" + "-36" \times "\frac{85}{3}" \end{bmatrix} - \begin{bmatrix} "-36" \times "-105" + "-36" \times "15" + + "4" \times "\frac{85}{3}" \end{bmatrix} \right) = \left(\frac{10060}{3} - [-1980] \right) = \frac{8000}{3}$ oe	M1A1
	Method 3	
	Finds the lengths: $DC = 4 - (-36) = 40$ and $EF = \frac{85}{3} - (-105) = \frac{400}{3}$	M1
	$\text{Area} = \frac{1}{2} \times '40' \times '\frac{400}{3}' = \frac{8000}{3}$ oe	M1A1
		Total 17 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 11 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 5 · find part c in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	Correct value of b. This may be embedded in a coordinate of B (10, 33)
	B1	For either $x = 4$ or $y = 15$
	B1	For (4, 15) or both $x = 4$ and $y = 15$
	ALT (Without finding the value of b)	
	B1	For $x = 4$ [by any method]
	B1	Substitutes their value of $x = 4$ into the given equation of l $y = 3\times'4'+3 = ...$
	B1	For $y = 15$
	M1	For identification of a gradient from the equation of l and use of $m = -\frac{1}{"3"}$
	dM1	For the correct use of any form of the equation of a line, if using $y = mx + c$, must reach $c = ...$ Note that $c = \frac{49}{3}$ Use of their (4, 15) which **must** have come from an attempt to find the midpoint of AB and their m. **Note:** Dependent upon the previous method mark.
	A1	$3y + x - 49 = 0$ accept terms on one side in any order. Cso **Note:** This equation is given. Award this and the previous mark that has come from work without any errors or omissions.
(b)	B1	Identifies the y intercept of line l (0, 3). This can be implied by later correct work or accept seen in part (a) Look for this coordinate written on a sketch if it is not in the body of the work.
	M1	Equates $-\frac{1}{"3"}$ to a **correct** expression for the gradient in terms of p and q, using **their** values of (0, 3) for the y intercept. **OR** for **correct** use of Pythagoras for the length of AD to form an equation in p and q, using **their** value of (0, 3) for the y intercept.
	dM1	For forming a quadratic equation in either p or q which must come from using their expression for the gradient and the length. **This is dependent on the first M mark in (b)**
	ddM1	For solving their quadratic equation by any acceptable method. **This is dependent on the previous two M marks in (b)**
	A1	For a correct p **or** q.
	A1	For a correct p **and** q.
	ALT	
	B1	Identifies the y intercept of line l (0, 3) This can be implied by later correct work or accept seen in part (a) Look for this coordinate written on a sketch if it is not in the body of the work.
	M1	Uses a correct method, scaling the gradient of $-\frac{1}{3}$ Must arrive at a scaling value (shown as a in the mark scheme), but it may not be the correct value. This mark may be implicit in work that follows.
	dM1	Uses their scaling factor, with the coordinates of A, to find a value for p or q **This is dependent on the first M mark in (b)**
	dM1	Uses their scaling factor, with the coordinates of A, to find a value for p and q **This is dependent on the first M mark in (b)**
	A1 A1	First A1 for p **or** q Second A1 for p **and** q
(c)	M1	Substitutes their value for – 36 into the equation for line k to find the y coordinate of point E
	M1	Substitutes their value for – 36 into the equation for line l to find the y coordinate of point F

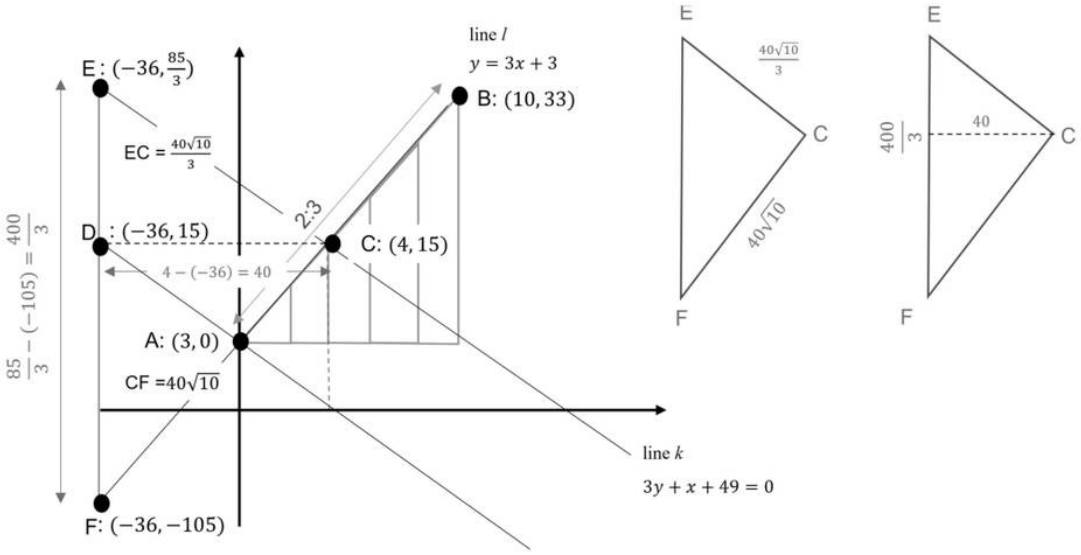
Type MS · Area of a triangle or quadrilateral from coordinates

Answer 11 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 4 of 5 · find part c in this mark scheme extract

	Method 1	
	M1	Uses Pythagoras with **their** values to find the length of CE **and** CF correctly.
	M1	Correct method to find the area of the triangle, using **their** values.
	A1	For $\frac{8000}{3}$ oe
	Method 2	
	M1	For a correct array using their values.
	M1	Correct calculation shown, using their values, takes the modulus and multiplies by $\frac{1}{2}$
	A1	$\frac{8000}{3}$ oe
	Method 3	
	M1	For finding the lengths DC **and** EF correctly using **their** values.
	M1	Applies the correct formula for the area of a triangle using their values.
	A1	For $\frac{8000}{3}$ oe

USEFUL SKETCH



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 11 of 17 · Question 11 part c · 5 marks

2022 JUN · Paper 01 (regional R) · extract 5 of 5 · find part c in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 12 of 17 · Question 10 part e · 3 marks

2025 NOV · Paper 01 · extract 1 of 4 · find part e in this mark scheme extract

Question	Scheme	Marks
10 (a)	$10-3a+a=0 \Rightarrow 10=2a \Rightarrow a=5$ *	M1A1cso [2]
(b)	$(0,-5)$	B1 [1]
(c) (i)	$A \quad (5,10)$ $C \quad (50,-5)$ $y=3x-5 \Rightarrow m_{L1}=3 \qquad m_{L2}=\dfrac{-5-10}{50-5}=\dfrac{-1}{3}$ $m_{L2}\times m_{L1}=\dfrac{-1}{3}\times 3=-1$	B1 M1A1 [3]
(ii)	$y-(-5)=\dfrac{-1}{3}(x-50) \Rightarrow y=\dfrac{-x}{3}+\dfrac{35}{3}$	M1A1 [2]
(d)	$-3=\dfrac{n-("-5")}{m} \Rightarrow n=-5-3m$ $2250=(50-m)^2+(-5-n)^2$ $2250=(50-m)^2+(-5-(-5-3m))^2=(50-m)^2+(3m)^2$ $2250=2500-100m+10m^2 \Rightarrow 0=25-10m+m^2=(m-5)^2$ $\Rightarrow m=5$ $\Rightarrow n=-5-3\times 5=-20$ **ALT** $-3=\dfrac{n-("-5")}{m} \Rightarrow m=\dfrac{-(n+5)}{3}$ $2250=(50-m)^2+(-5-n)^2$ $2250=\left(50+\dfrac{n+5}{3}\right)^2+n^2+10n+25$ $2250=2500+\dfrac{100(n+5)}{3}+\dfrac{n^2+10n+25}{9}+n^2+10n+25$ $\Rightarrow \quad 0=9(2500-2250)+300(n+5)+10n^2+100n+250$ $=2500+300n+1500+10n^2+100n$ $=n^2+40n+400$ $=(n+20)(n+20)$ $\Rightarrow n=-20$ $\Rightarrow m=\dfrac{-(-20+5)}{3}=5$	M1 M1 M1 M1 A1 A1 [6] [M1] [M1] [M1] [M1] [A1] [A1]
(e)	Correctly uses the formula for a kite using their lengths	M1

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 12 of 17 · Question 10 part e · 3 marks

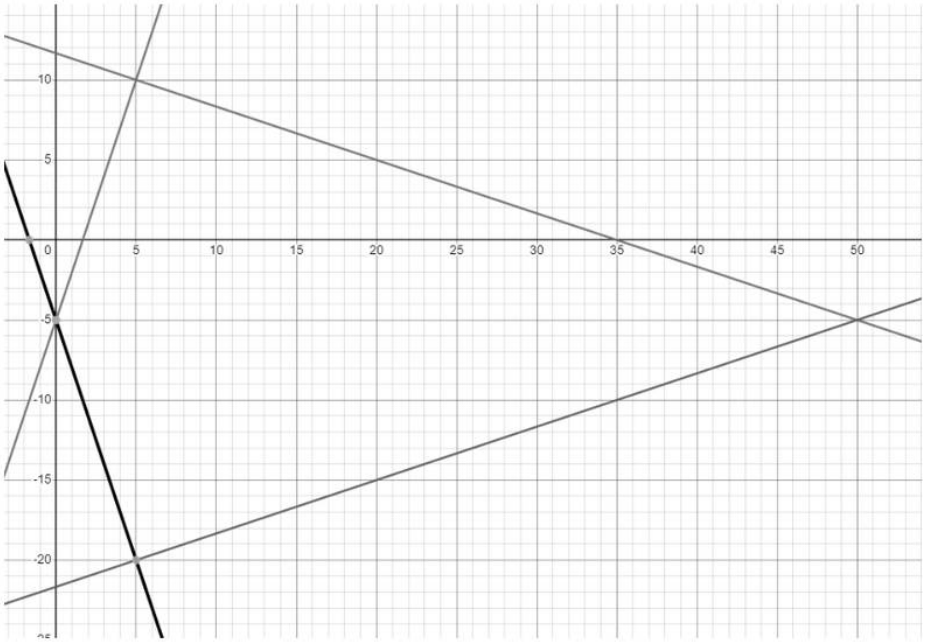
2025 NOV · Paper 01 · extract 2 of 4 · find part e in this mark scheme extract

	$\text{Area} = \frac{50\times(10-(-20))}{2} = 750$	dM1A1 [3]
	ALT 1 Correctly uses determinant method with their co-ordinates $\text{Area} = \frac{1}{2}\begin{pmatrix} 0 & 5 & 50 & 5 & 0 \\ -5 & 10 & -5 & -20 & -5 \end{pmatrix} = 750$	[M1 dM1A1]
	ALT 2 Correctly uses formulae for two triangles ABD and ACD using their lengths $\text{Area} = 2\times\left(\frac{1}{2}\times 15\times 50\right) = 750$ Or $\text{Area} = \left(\frac{1}{2}\times 30\times 5\right)+\left(\frac{1}{2}\times 45\times 30\right) = 750$	[M1 dM1A1]
		Total 17 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 12 of 17 · Question 10 part e · 3 marks

2025 NOV · Paper 01 · extract 3 of 4 · find part e in this mark scheme extract



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 12 of 17 · Question 10 part e · 3 marks

2025 NOV · Paper 01 · extract 4 of 4 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 13 of 17 · Question 4 part e · 2 marks

2021 JUN · Paper 02 · extract 1 of 5 · find part e in this mark scheme extract

number		
4 (a)	Gradient $= \frac{11+10}{3+4} = 3$	
	$y+10=3(x+4)$ or $y-11=3(x-3)$ oe	M1A1 (2)
(b)	e.g. $\left(\frac{4\times-4+3\times3}{3+4}, \frac{4\times-10+3\times11}{3+4}\right)=(-1,-1)$	M1 A1 (2)
ALT (b)	Using Vectors	
	$\begin{pmatrix}-4\\-10\end{pmatrix}+\frac{3}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$ or $\begin{pmatrix}3\\11\end{pmatrix}-\frac{4}{7}\begin{pmatrix}7\\21\end{pmatrix}=\begin{pmatrix}-1\\-1\end{pmatrix}$	{M1} {A1}
(c)	$-\frac{1}{3}=\frac{n+1}{m+1} \Rightarrow -\frac{1}{3}(m+1)=n+1$	M1
	$\left(\sqrt{10}\right)^2=(m+1)^2+(n+1)^2$	M1
	$10=(m+1)^2+\frac{1}{9}(m+1)^2$	M1
	$9=(m+1)^2$	M1
	$m=-4 \quad n=0$	A1 A1 (6)
ALT (c)	Using Vectors	
	$\overrightarrow{AB}=\begin{pmatrix}7\\21\end{pmatrix}$ so perpendicular to $\overrightarrow{AB}=\begin{pmatrix}21\\-7\end{pmatrix}$	{M1}
	$\left\lvert\overrightarrow{AB}\right\rvert=7\sqrt{10} \Rightarrow, \left\lvert\overrightarrow{AP}\right\rvert=3\sqrt{10}$	{M1,M1}
	$\overrightarrow{PQ}=\frac{\sqrt{10}}{7\sqrt{10}}\times\begin{pmatrix}21\\-7\end{pmatrix}=\begin{pmatrix}3\\-1\end{pmatrix}$	{M1}
	So $Q=(-1-3,-1--1)$	{A1}
	$Q=(-4,0)$	{A1}
(d)(i)	$AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$	M1
	$RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$	A1
(d)(ii)	Gradient of $RQ=\frac{-21-0}{-11+4}=3$	M1
	So Gradient of AB (=3) = Gradient of RQ	A1 (4)

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 13 of 17 · Question 4 part e · 2 marks

2021 JUN · Paper 02 · extract 2 of 5 · find part e in this mark scheme extract

ALT (d)	Using Vectors	
	$\overrightarrow{RQ} = \begin{pmatrix} -4-(-11) \\ 0-(-21) \end{pmatrix} = \begin{pmatrix} 7 \\ 21 \end{pmatrix}$	{M1} {A1}
	$\overrightarrow{AB} = \begin{pmatrix} 7 \\ 21 \end{pmatrix} = \overrightarrow{RQ}$	{M1}
	Because the vectors are the same they must be parallel and the same length	{A1}
(e)	Area $= 7\sqrt{10} \times \sqrt{10} = 70$	M1 A1 (2)
ALT (e)	Using Vectors	
	$\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & -4 & 3 \\ 11 & 10 & -21 & 0 & 11 \end{vmatrix}$	{M1}
	$= 70$	{A1}
		Total is 16 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 13 of 17 · Question 4 part e · 2 marks

2021 JUN · Paper 02 · extract 3 of 5 · find part e in this mark scheme extract

Part	Mark	Guidance
(a)	M1	For a fully correct method of finding an equation of a straight line. $\frac{y-y_1}{y_2-y_1}=\frac{x-x_1}{x_2-x_1}\Rightarrow\frac{y-(-10)}{11-(-10)}=\frac{x-(-4)}{3-(-4)}$ Or finds gradient $\frac{11+10}{3+4}=3$ and uses $y+10=3(x+4)$ or $y-11=3(x-3)$ If $y=mx+c$ is used, they must find a complete equation for this mark. Allow **one** error only for the award of this mark.
	A1	For a correct line in any form. $y+10=3(x+4)$ or $y-11=3(x-3)$ or $y=3x+2$ or even $\frac{y+10}{21}=\frac{x+4}{7}$ but do not allow incomplete processing.
(b)	M1	For one correct from $x=-1$ or $y=-1$
	A1	For the correct coordinates of point P $(-1,-1)$ Accept $x=-1$ $y=-1$
(c)	M1	Uses the perpendicular gradient to set up an equation in m and n. $-\frac{1}{'3'}=\frac{n-'(-1)'}{m-'(-1)'}\Rightarrow-\frac{1}{'3'}(m+1)=n+1$ or $n=-\frac{1}{3}m-\frac{4}{3}$ Ft their gradient in part (a) and their P from part (b) for this mark.
	M1	Uses Pythagoras theorem to set up an equation in m and n. $\left(\sqrt{10}\right)^2=(m-'(-1)')^2+(n-'(-1)')^2$ Ft their coordinates of point P form part (b) for this mark.
	M1	Attempts to solve their two equations in n and m simultaneously and forms a quadratic equation in one variable only. $10=(m+1)^2+\frac{1}{9}(m+1)^2\Rightarrow 9=(m+1)^2$ or $0=m^2+2m-8$ or $10=9(n+1)^2+(n+1)^2\Rightarrow 0=10n^2+20n$
	M1	For solving their either: $9=(m+1)^2\Rightarrow m=...$ or $0=10n^2+20n\Rightarrow n=...$ which must be a quadratic equation.
	A1	For finding **either** $m=-4$ **or** $n=0$ Condone the sight of $m=2$ for this mark.
	A1	For finding **both** $m=-4$ **and** $n=0$ $\Rightarrow(-4,\ 0)$ The final answer must be given as coordinates.
	ALT – using vectors – see main scheme.	
(d)(i)	M1	For finding **either** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **Or** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$
	A1	For finding **both** the length $AB=\sqrt{(3+4)^2+(11+10)^2}=7\sqrt{10}$ **And** $RQ=\sqrt{(-11+4)^2+(-21)^2}=7\sqrt{10}$ **and** states they are equal
(d)(ii)	M1	The gradient of $RQ=\frac{-21-'0'}{-11-'(-4)'}='3'$ Ft their coordinates from part (c)

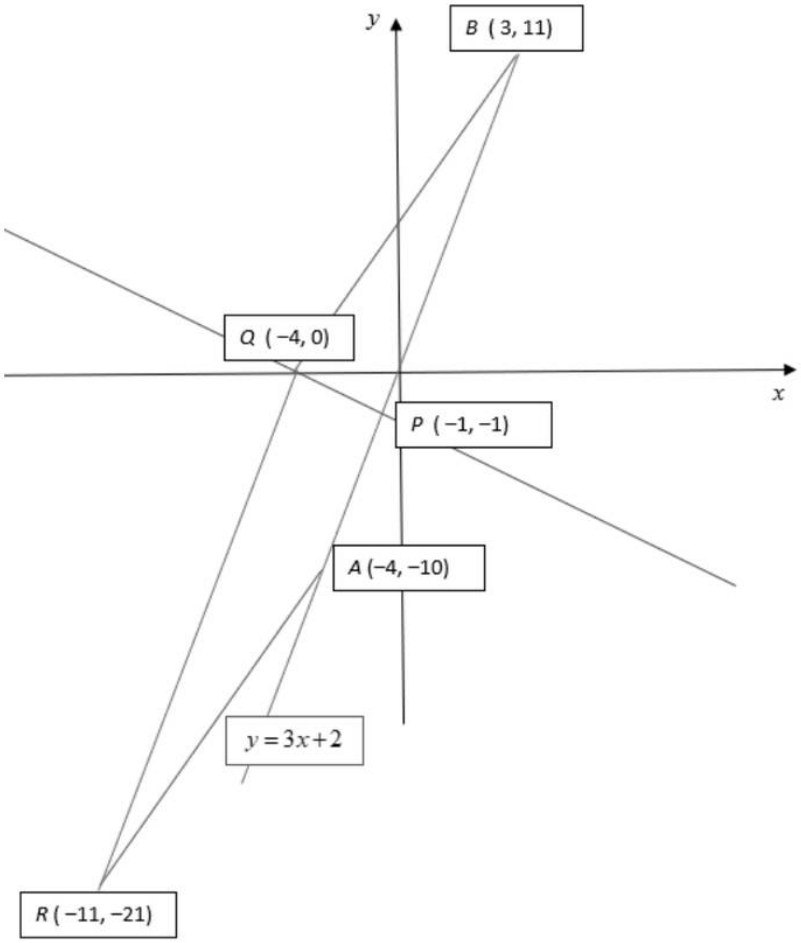
Type MS · Area of a triangle or quadrilateral from coordinates

Answer 13 of 17 · Question 4 part e · 2 marks

2021 JUN · Paper 02 · extract 4 of 5 · find part e in this mark scheme extract

	A1	States that the gradient of RQ = gradient of AB [from (a)]
	ALT – Uses vectors, see main scheme. Ft their coordinates of Q – (m, n)	
(e)	**M1**	For a correct expression for the area using their length of AB and the given length of PQ $\left(\sqrt{10}\right)$ Area = $'7\sqrt{10}'\times\sqrt{10}=...$
	A1	For the area = 70 [square units]
	ALT – Uses the discriminant	
	M1	For a correct expression of the area in sequential order using their coordinates for Q Area = $\frac{1}{2}\begin{vmatrix} 3 & -4 & -11 & '-4' & 3 \\ 11 & 10 & -21 & '0' & 11 \end{vmatrix}$
	A1	Area = 70 [square units]

Useful sketch



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 13 of 17 · Question 4 part e · 2 marks

2021 JUN · Paper 02 · extract 5 of 5 · find part e in this mark scheme extract

Question number	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 14 of 17 · Question 8 part e · 4 marks

2023 JUN · Paper 02 (regional R) · extract 1 of 5 · find part e in this mark scheme extract

8(a)	$\frac{y-5}{5-9}=\frac{x-1}{1-9}$ $\Rightarrow x-2y+9=0$	M1A1 A1 [3]
(b)	Coordinates of point X $\left(\frac{3\times9+1\times1}{3+1},\frac{3\times9+1\times5}{3+1}\right)=(7,8)$ The perpendicular gradient $=-\frac{1}{\frac{1}{2}}=-2$ Equation of l $y-8=-2(x-7)\Rightarrow y=-2x+22$ *	B1B1 B1ft M1A1 cso [5]
(c)	$y=-2(6)+22\Rightarrow p=10$	B1 [1]
(d)	$\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow \overrightarrow{CD}=\begin{pmatrix}6-8\\10-4\end{pmatrix}=\begin{pmatrix}-2\\6\end{pmatrix}$ and coordinates of C are (6, 10) so coordinates of D are $(-2, 6)$	M1 A1A1 [3]
(e)	Length of AB (or CD) $=\sqrt{(9-5)^2+(9-1)^2}=\sqrt{80}$ Length of $CX=\sqrt{(10-8)^2+(7-6)^2}=\sqrt{5}$ Area of parallelogram $ABCD=\sqrt{5}\times\sqrt{80}=\sqrt{400}=20$ (units^2)	B1 B1 M1A1 [4]
		Total 16 marks

Part	Mark	Notes
(a)	M1	For using a correct method and the given coordinates of A and B to form the equation of AB. Do not score this mark until they find the gradient and form the equation of the line using a correct formula. If they use $y=mx+c$ do not allow this mark until they find c and form a complete equation. $\left[m=\frac{1}{2},\quad c=\frac{9}{2}\quad y=\frac{x}{2}+\frac{9}{2}\right]$
	A1	For the correct equation of AB in any form.
	A1	For the correct equation of AB in the required form.

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 14 of 17 · Question 8 part e · 4 marks

2023 JUN · Paper 02 (regional R) · extract 2 of 5 · find part e in this mark scheme extract

(b)	B1	For either x or y correct coordinates of point X **NB This is an M mark in Epen**
	B1	For both x and y correct coordinates of point X **NB This is an A mark in Epen.**
	B1ft	For writing down the inverse reciprocal of their gradient for AB Ft their gradient form part (a)
	M1	For forming an equation for l using their coordinates of X and their negative reciprocal gradient of AB $y-8=-2(x-7)\Rightarrow y=-2x+22$
	A1	For the correct equation of l as shown.
(c)	B1	For $y = 10$
(d)	M1	For a suitable method. **Method 1 - Uses vectors:** $\overrightarrow{BA}=\begin{pmatrix}1-9\\5-9\end{pmatrix}=\begin{pmatrix}-8\\-4\end{pmatrix}\Rightarrow\overrightarrow{CD}=\begin{pmatrix}6+(-8)\\10+(-4)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ OR: $\overrightarrow{BC}=\begin{pmatrix}6-9\\9-10\end{pmatrix}=\begin{pmatrix}-3\\-1\end{pmatrix}\Rightarrow\overrightarrow{AD}=\begin{pmatrix}1-3\\5-(-1)\end{pmatrix}\Rightarrow\left[\begin{pmatrix}-2\\6\end{pmatrix}\right]$ **Method 2 - Uses simultaneous equations**: The equation of AD is $y=-\frac{x}{3}+\frac{16}{3}$ and of CD is $y=\frac{x}{2}+7$ $-\frac{x}{3}+\frac{16}{3}=\frac{x}{2}+7\Rightarrow x=-2$ and $y=6$ **Method 3 - Uses the gradient and length of *AB* or *CD*** $AB=CD=4\sqrt{5}=\sqrt{(6-x)^2+(y-10)^2}$ Gradient: $\frac{1}{2}=\frac{10-y}{6-x}\Rightarrow x=2y-14$ $\left(4\sqrt{5}\right)^2=(6-(2y-14))x^2+(y-10)^2\Rightarrow 5y^2-100y+420=0$ $\Rightarrow y=6,14$ and $x=-2,14$ Allow no more than one error in either method.
	A1	For either correct x or y coordinate of D [(– 2, 6)] **NB** this is an A mark in Epen
	A1	For both correct coordinates of D (– 2, 6)

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 14 of 17 · Question 8 part e · 4 marks

2023 JUN · Paper 02 (regional R) · extract 3 of 5 · find part e in this mark scheme extract

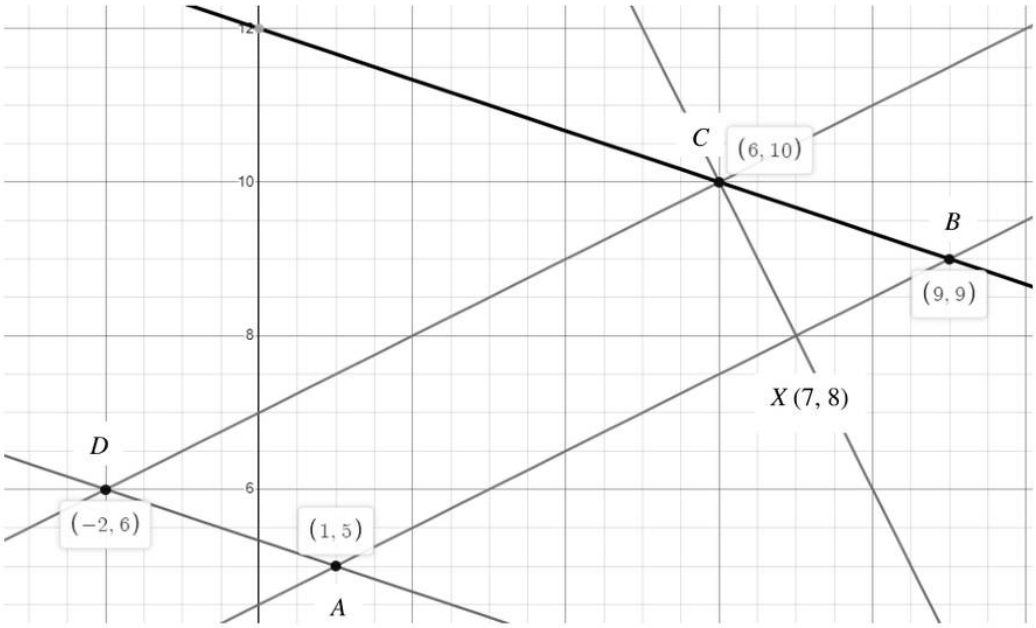
(e)	B1	For the correct length of either ***AB*** **[*CD*] or *CX*** These are given coordinates there is no ft
	B1	For the correct lengths of both ***AB*** **[*CD*] and *CX*** Ft their C and their X
	M1	For using a correct method to calculate the area of a parallelogram. $\text{b} \times \text{h} = CX \times AB = \sqrt{5} \times \sqrt{80} = \sqrt{400} = 20$ If their lengths are incorrect allow this mark provided their lengths are identified as base and perpendicular height. If they use AD as the base $\left[\sqrt{10}\right]$, the perpendicular height required is $2\sqrt{10}$ so base × height = $\sqrt{10} \times 2\sqrt{10} = ...$
	A1	For the correct area of 20 [units^2]
	ALT – Using determinants	
	B1B1	For sight of the correct array which must use the coordinates of A, B, C and D **ONLY.** The coordinates (7, 8) seen in the array is B0B0 $\begin{pmatrix} 1 & 9 & '6' & '-2' & 1 \\ 5 & 9 & '10' & '6' & 5 \end{pmatrix}$ Award both marks for fully correct. Award B1 if there are no more than 2 errors but with none missing. There must be 5 sets of coordinates in the array with first and last the same. **The coordinates must go in order around the parallelogram clockwise or anticlockwise.**
	M1	For the correct evaluation of their 2×5 array. Allow this even if they have (7, 8) included instead of (9, 9) [which is a common error] $\frac{1}{2}\begin{pmatrix} 1 & 9 & 6 & -2 & 1 \\ 5 & 9 & 10 & 6 & 5 \end{pmatrix} = \frac{1}{2}\left\|\left[(9+90+36-10)-(45+54-20+6)\right]\right\| = ...$
	A1	For the correct area of 20 [units^2]

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 14 of 17 · Question 8 part e · 4 marks

2023 JUN · Paper 02 (regional R) · extract 4 of 5 · find part e in this mark scheme extract

Useful sketch



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 14 of 17 · Question 8 part e · 4 marks

2023 JUN · Paper 02 (regional R) · extract 5 of 5 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 15 of 17 · Question 11 part d.ii · 4 marks

2021 NOV · Paper 02 · extract 1 of 3 · find part d.ii in this mark scheme extract

Number		
11 (a)	$\frac{y-5}{5--5}=\frac{x-7}{7--3}$ $[\Rightarrow y=x-2]$ oe	M1A1 [2]
(b)	$m_{BC}=\frac{5-1}{7-p}=\frac{4}{7-p}$, $m_{AC}=\frac{1-(-5)}{p-(-3)}=\frac{6}{p+3}$ $m_{BC}\times m_{AC}=-1\Rightarrow\frac{4}{7-p}\times\frac{6}{p+3}=-1\Rightarrow\frac{4}{7-p}=-\left(\frac{p+3}{6}\right)$ $\Rightarrow -24=-p^2+4p+21\Rightarrow p^2-4p-45=0$ $\Rightarrow (p-9)(p+5)=0\Rightarrow p=-5,\ (9)$	M1A1A1 M1 dM1 M1A1* [7]
(c)	Using formula; $\left(\frac{(7\times-5)-(3\times7)}{7-3},\frac{(7\times1)-(3\times5)}{7-3}\right)=(-14,-2)$	B1B1 [2]
(d) (i)	$AC=\sqrt{(1-[-5])^2+(-5-[-3])^2}=\sqrt{40}=2\sqrt{10}$	B1
(ii)	$BD=\sqrt{(7-[-14])^2+(5-[-2])^2}=\sqrt{490}=7\sqrt{10}$ Area $=\frac{1}{2}\times2\sqrt{10}\times7\sqrt{10}=70\ (\text{units}^2)$	M1 M1A1 [4]
		Total 15 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 15 of 17 · Question 11 part d.ii · 4 marks

2021 NOV · Paper 02 · extract 2 of 3 · find part d.ii in this mark scheme extract

Part	Mark	Notes
(a)	M1	For forming an equation of a straight line
	A1	Accept a correct line in any form following simplification of denominators; for example, accept $\frac{y-5}{10}=\frac{x-7}{10}$ or $y-5=x-7$ or $y+5=x+3$
(b)	M1	For attempting to find an expression for the gradient of BC **or** the gradient of AC $m_{BC}=\frac{5-1}{7-p}$ or $m_{AC}=\frac{1-(-5)}{p-(-3)}$ Accept maximum one slip in either expression for this mark.
	A1	For an expression for the gradient of BC **or** AC fully correct
	A1	For an expression for the gradient of BC **and** AC fully correct Need not be simplified in either case
	M1	For setting their gradient of BC = negative reciprocal of their gradient of AC
	dM1	For attempting to form a 3TQ in p: $\left[p^2-4p-45=0\right]$ Dependent on the previous M mark.
	M1	For solving their 3TQ by any method. See General Guidance.
	A1*	For $p=-5$
(c)	B1	For either $x=-14$ or $y=-2$
	B1	Both coordinates correct $(-14,-2)$ given as coordinates only.
(d)(i)	B1	For the correct length of AC $\left[=2\sqrt{10}\right]$
(ii)	M1	For the correct method to find length BD $\left[=7\sqrt{10}\right]$ Ft their $x=-14$ or $y=-2$
	M1	For a correct method using $A=\frac{1}{2}\times\text{base}\times\text{height}$ using their AC and BD
	A1	For the correct area of $70\ (\text{units}^2)$
	ALT – Uses determinants	
	M1	For stating $A=\frac{1}{2}\begin{pmatrix}-3 & 7 & -14 & -3\\ -5 & 5 & -2 & -5\end{pmatrix}$ oe
	M1	For evaluating their determinant $A=\frac{1}{2}\left(\left[-3\times5+7\times-2+-14\times-5\right]-\left[7\times-5+-14\times5+-3\times-2\right]\right)$ oe
	A1	Area = $70\ (\text{units}^2)$ Which must be a positive value

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 15 of 17 · Question 11 part d.ii · 4 marks

2021 NOV · Paper 02 · extract 3 of 3 · find part d.ii in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 16 of 17 · Question 8 part e · 3 marks

2023 JUN · Paper 02 · extract 1 of 5 · find part e in this mark scheme extract

Question	Scheme	Marks
8(a)	$(\text{Grad}_{AB}) = \frac{2-8}{12-(-6)}$ oe $y-2="\left(-\frac{1}{3}\right)"(x-12)$ oe $\Rightarrow x+3y-18=0$ or $-x-3y+18=0$ oe	M1 dM1 (A1 on ePen) A1 [3]
(b)	Length $=\sqrt{(12--6)^2+(2-8)^2}=6\sqrt{10}$ eg $\sqrt{360}$ oe	M1A1 [2]
(c)	$\left(\left(\frac{2\times-6+1\times12}{1+2}, \frac{2\times8+1\times2}{1+2}\right)\right)$ $(0,6)$ or $m=0, n=6$	B1 B1 (M1A1 on ePen) [2]
(d) Mark parts (i) and (ii) together	Gradient of $CA=\frac{q-8}{p+6}$ **or** gradient of $CB=\frac{q-2}{p-12}$ oe $\frac{q-8}{p+6}=-\frac{1}{\frac{q-2}{p-12}}=-\left(\frac{p-12}{q-2}\right)\Rightarrow q^2-10q+16=-p^2+6p+72$ Gradient of $XC=-\frac{1}{"-\frac{1}{3}"}\Rightarrow-\frac{1}{"-\frac{1}{3}"}=\frac{q-6}{p-0}(\Rightarrow q=3p+6)$ $('3p+6')^2-10('3p+6')+16=-p^2+6p+72\Rightarrow 10p^2-80=0$ $10p^2-80=0\Rightarrow p=\sqrt{8}$ oe $q=3\times\sqrt{8}+6=6+6\sqrt{2}$ oe	M1 M1 B1ft ddM1A1 M1A1 [7]
ALT1 Mark parts (i) and (ii) together	(midpoint of AB $=)\left(\frac{-6+12}{2}, \frac{8+2}{2}\right)$ $(=(3,5))$ (radius of C) $=\frac{"6\sqrt{10}"}{2}$ oe or $\sqrt{(-6-3)^2+(8-5)^2}$ oe $y=\frac{-1}{"-\frac{1}{3}"}x+6$ $(y=3x+6)$ oe $(x-"3")^2+(y-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2\Rightarrow(x-"3")^2+("3x+6"-"5")^2=\left(\frac{"6\sqrt{10}"}{2}\right)^2$ $10x^2=80$ oe $x=\sqrt{8}$ oe $p=\sqrt{8}, q=6+3\sqrt{8}$ oe	M1 M1 B1ft ddM1 A1 M1 A1 [7]

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 16 of 17 · Question 8 part e · 3 marks

2023 JUN · Paper 02 · extract 2 of 5 · find part e in this mark scheme extract

ALT2 Mark parts (i) and (ii) together	$\left((AC)^2=\right)(p--6)^2+(q-8)^2$ or $(AC=)\sqrt{(p--6)^2+(q-8)^2}$ oe	M1
	$\left((BC)^2=\right)(p-12)^2+(q-2)^2$ or $(AC=)\sqrt{(p-12)^2+(q-2)^2}$ oe	M1
	$q=3p+6$	B1ft
	$\left((AB)^2=(AC)^2+(BC)^2\right)$	
	$\left("6\sqrt{10}"\right)^2=(p--6)^2+("3p+6"-8)^2+(p-12)^2+("3p+6"-2)^2$	ddM1A1
	$10p^2=80$ oe	
	$p=\sqrt{8}$ oe	M1 A1
	$p=\sqrt{8}, q=6+3\sqrt{8}$ oe	[7]
(e)	Length $CX=\sqrt{\left("\sqrt{8}"-"0"\right)^2+\left("6+6\sqrt{2}"-"6"\right)^2}\left(=4\sqrt{5}\right)$	M1
	Area of triangle $ABC=\frac{1}{2}\times"4\sqrt{5}"\times"6\sqrt{10}"=60\sqrt{2}$ oe	dM1 A1 [3]
ALT	$\frac{1}{2}\begin{vmatrix}-6 & 12 & "\sqrt{8}" & -6\\ 8 & 2 & "6+3\sqrt{8}" & 8\end{vmatrix}$ oe	M1
	$\frac{1}{2}\left[\left(-6\times2+12\times"(6+3\sqrt{8})"+"\sqrt{8}"\times8\right)-\left(-6\times"(6+3\sqrt{8})"+"\sqrt{8}"\times2+12\times8\right)\right]$ oe	dM1
	$60\sqrt{2}$ oe	A1 [3]
		Total 17 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 16 of 17 · Question 8 part e · 3 marks

2023 JUN · Paper 02 · extract 3 of 5 · find part e in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correctly finding the gradient of AB in unsimplified form.
	dM1 (A1 on ePen)	For a full and correct attempt to find the equation of the line using their gradient. No simplification is required. If using $y = mx + c$, a value for c must be found.
	A1	For a correct equation in the required form.
(b)	M1	For using a correct method to find the length of line segment AB, in unsimplified form.
	A1	For the correct exact length.
(c)	B1 (M1 on ePen)	For either coordinate correct.
	B1 (A1 on ePen)	For both correct coordinates (0, 6) For part c the values of m and n can be explicitly identified or written in a coordinate.
In part (d) allow p to be interchangeable with x, q to be interchangeable with y throughout		
(d)	M1	For a correct statement of the gradient for either CA or CB
	M1	For using the negative perpendicular of either gradient and equating the gradients to form an equation in terms of p and q only.
	B1ft	For finding the negative reciprocal of their gradient of XC and placing this equal to a correct expression in p and q as shown.
	ddM1	For correctly substituting their **linear** expression for p or q into a **quadratic** equation in q or p to obtain an equation in one variable. Must use their $q = 3p + 6$ and is dependent on both previous method marks.
	A1	For the correct two term quadratic..
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.
ALT1 (d)	M1	For the correct method to find the midpoint of AB
	M1	For the correct method to find the radius of C, ft their answer from part b if used.
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their $y = 3x + 6$ into the equation of a circle, using their midpoint of AB and their radius of C Must use their $y = 3x + 6$ and is dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either x or y
	A1	For **both** p and q correct.
ALT2 (d)	M1	For the correct method to find the length of AC or $(AC)^2$
	M1	For the correct method to find the length of BC or $(BC)^2$
	B1ft	For the equation of the line, unsimplified, ft their gradient of AB
	ddM1	For correctly substituting their lengths and their $q = 3p + 6$ into a correct Pythagorean equation, dependent on both previous method marks.
	A1	Correct equation
	M1	For **correctly solving their** quadratic to find a value for either p or q
	A1	For **both** p and q correct.

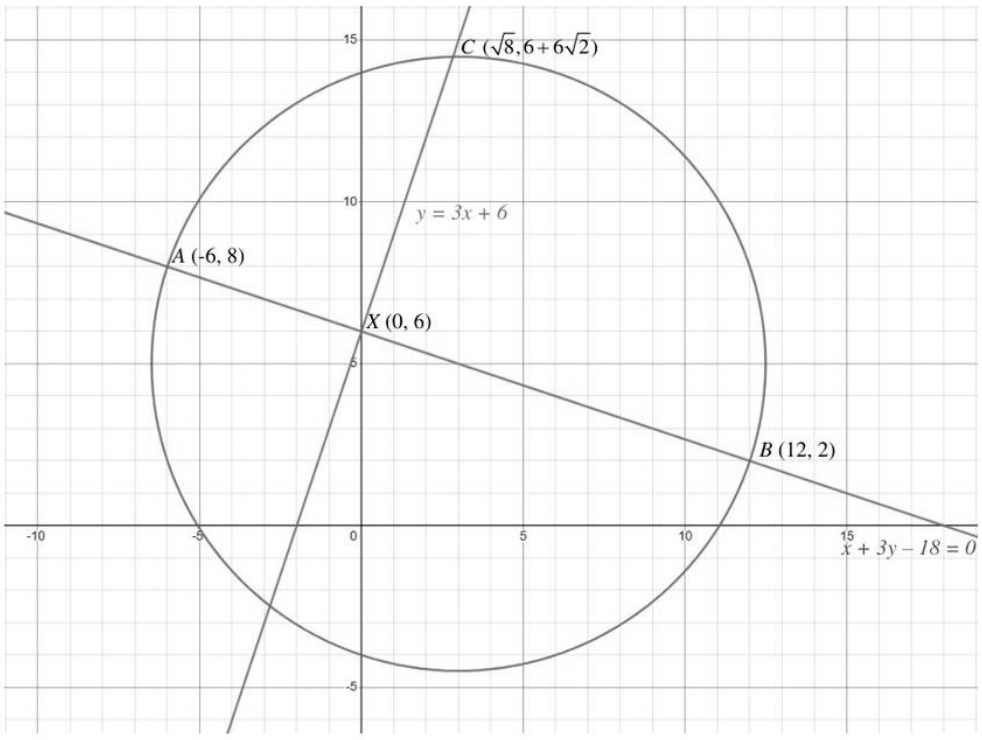
Type MS · Area of a triangle or quadrilateral from coordinates

Answer 16 of 17 · Question 8 part e · 3 marks

2023 JUN · Paper 02 · extract 4 of 5 · find part e in this mark scheme extract

(e)	**M1**	For using a correct method to find the length of the perpendicular from AB to C
	dM1	For using their results from part (b) and their length of perpendicular from AB to C with the correct formula for the area of a triangle. Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)
ALT	**M1**	For a correct statement for the area such as the one shown, using their values of p and q
	dM1	For the correct evaluation of their determinant Dependent on previous method mark.
	A1	For the correct area of $60\sqrt{2}$ or $30\sqrt{8}$ (units^2)

Useful Sketch for Parts c/d – look for any working on or near a sketch.



Type MS · Area of a triangle or quadrilateral from coordinates

Answer 16 of 17 · Question 8 part e · 3 marks

2023 JUN · Paper 02 · extract 5 of 5 · find part e in this mark scheme extract

Question	Scheme	Marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 17 of 17 · Question 3 part b · 5 marks

2022 JAN · Paper 01 (regional R) · extract 1 of 4 · find part b in this mark scheme extract

Number		
3 (a)	(Midpoint of AB =) $(5,5)$	B1
	(Gradient of AB =) $\frac{3-7}{9-1}=\left(-\frac{1}{2}\right)$ oe	M1
	(Gradient of l =) $-\left(\frac{1}{"-\frac{1}{2}"}\right)$ (= 2)	A1ft
	$y-"5"="2"(x-"5")$ oe	M1 A1 (5)
ALT	Let D (x, y) be any point on the line	
	$x-1^{\ 2}+\ y-7^{\ 2}=\ x-9^{\ 2}+\ y-3^{\ 2}$	{B1}
	$x^2-2x+1+y^2-14y+49=x^2-18x+81+y^2-6y+9$	{M1} {A1ft}
	$16x-40=8y \rightarrow (y=2x-5)$	M1 A1
(b)	$x=\frac{5}{2}$	B1ft
	(length of C to midpoint of AB =) $\sqrt{("5"-0)^2+("5"-2.5)^2}\left(=\frac{5\sqrt{5}}{2}\right)$	M1
	(length of AB =) $\sqrt{8^2+4^2}\left(=4\sqrt{5}\right)$	B1 (A1 on ePen)
	$\frac{1}{2}\times4\sqrt{5}\times"\frac{5\sqrt{5}}{2}"$	M1
	25	A1 [5]
ALT1	Use of determinant	
	$x=\frac{5}{2}$	B1ft
	$\frac{1}{2}\begin{vmatrix}1 & 9 & "2.5" & 1\\ 7 & 3 & 0 & 7\end{vmatrix}$	M1 A1
	$\left(\frac{1}{2}\right)\left\|(1\times3+9\times0+"2.5"\times7)-(1\times0+"2.5"\times3+9\times7)\right\|$ oe	M1
	25	A1 [5]

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 17 of 17 · Question 3 part b · 5 marks

2022 JAN · Paper 01 (regional R) · extract 2 of 4 · find part b in this mark scheme extract

ALT2	Area of rectangle – (sum of areas of 3 triangles)	
	$x = \frac{5}{2}$	B1ft
	$\frac{1}{2}\times\left("\frac{5}{2}"-1\right)\times 7+\frac{1}{2}\times\left(9-"\frac{5}{2}"\right)\times 3+\frac{1}{2}(9-1)(7-3)$ $\left(=\frac{21}{4}+\frac{39}{4}+16=31\right)$	M1
	$(9-1)\times 7=56$	A1
	56 – "31"	M1
	35	A1 [5]
		Total 10 marks

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 17 of 17 · Question 3 part b · 5 marks

2022 JAN · Paper 01 (regional R) · extract 3 of 4 · find part b in this mark scheme extract

Part	Marks	Notes
(a)	B1	For midpoint of $AB = (5,5)$
	M1	For correctly finding the unsimplified gradient of $AB = \frac{3-7}{9-1}$ oe
	A1ft	For correctly finding the unsimplified gradient of line l, use of their gradient for AB to find the negative reciprocal. $-\left(\frac{1}{"-\frac{1}{2}"}\right)$
	M1	For a correct method to find the equation of l using their midpoint and their gradient, which must have involved finding the negative reciprocal of the gradient of AB
	A1	For $y-5=2(x-5)$ ft oe
ALT	B1	For correctly equating the two expressions for AD^2 and AB^2 to give the equation shown
	M1	For an expansion of either their LHS or their RHS provided their equation is in the form. $x-a^{\ 2}+\ y-b^{\ 2}=\ x-c^{\ 2}+\ y-d^{\ 2}$ Allow one numerical or algebraic slip.
	A1ft	For a correct expansion of both their LHS and their RHS provided their equation is in the form. $x-a^{\ 2}+\ y-b^{\ 2}=\ x-c^{\ 2}+\ y-d^{\ 2}$
	M1	For simplifying leading to a linear equation
	A1	For $y=2x-5$ oe
(b)	B1ft	For $x=\frac{5}{2}$ ft their equation from part (a) with $y=0$
	M1	For finding the unsimplified length of C to midpoint of AB = $\sqrt{("5"-0)^2+("5"-"2.5")^2}\left(=\frac{5\sqrt{5}}{2}\right)$ Allow use of their (5, 5) and their C
	B1 (A1 on ePen)	For finding the unsimplified length of $AB=\sqrt{8^2+4^2}\left(=4\sqrt{5}\right)$
	M1	For area of triangle = $\frac{1}{2}\times 4\sqrt{5}\times"\frac{5\sqrt{5}}{2}"$
	A1	For 25
ALT1	B1ft	As main
	M1	For the determinant set up with their C
	A1	For the determinant set up fully correctly
	M1	For correctly evaluating their determinant, with their C, full working must be shown.
	A1	25
ALT2	B1ft	As main
	M1	For the correct calculation of the three areas using their C
	A1	For the correct rectangle area

Type MS · Area of a triangle or quadrilateral from coordinates

Answer 17 of 17 · Question 3 part b · 5 marks

2022 JAN · Paper 01 (regional R) · extract 4 of 4 · find part b in this mark scheme extract

	M1	For the correct calculation, using their values for the three areas
	A1	25

Question number	Scheme	Marks